



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

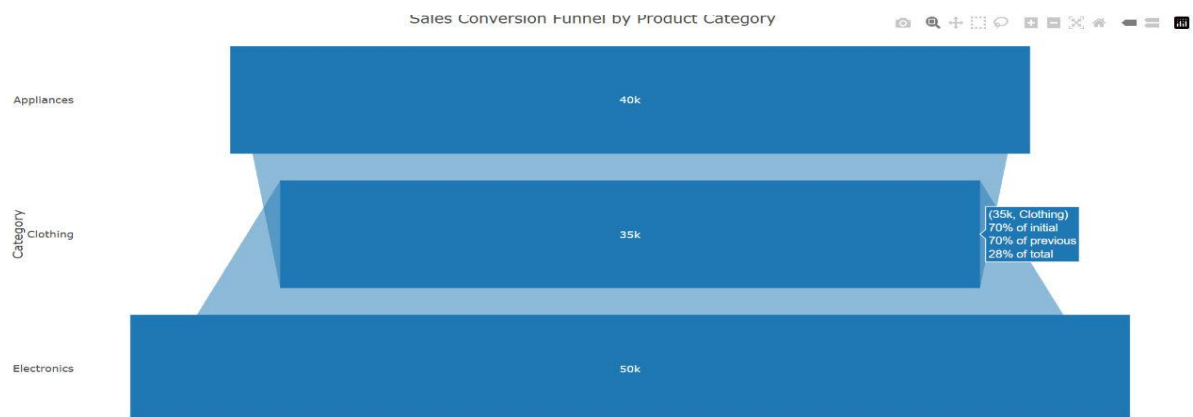
SET 11- Product Category Analysis

11 A. Funnel Chart

Program:

```
product_data <- data.frame(  
  Category=c("Electronics","Clothing","Appliances"),  
  Sales=c(50000,35000,40000))  
  
plot_ly( product_data,  
  type="funnel",  
  y=~Category,  
  x=~Sales) %>%  
  layout(title="Sales Conversion Funnel by Product Category")
```

Output:



11 B. Table

Program:

```
kable(  
  product_data,  
  caption="Product Category Sales")
```

Output:

Table: Product Category Sales

Category	Sales
Electronics	50000
Clothing	35000
Appliances	40000

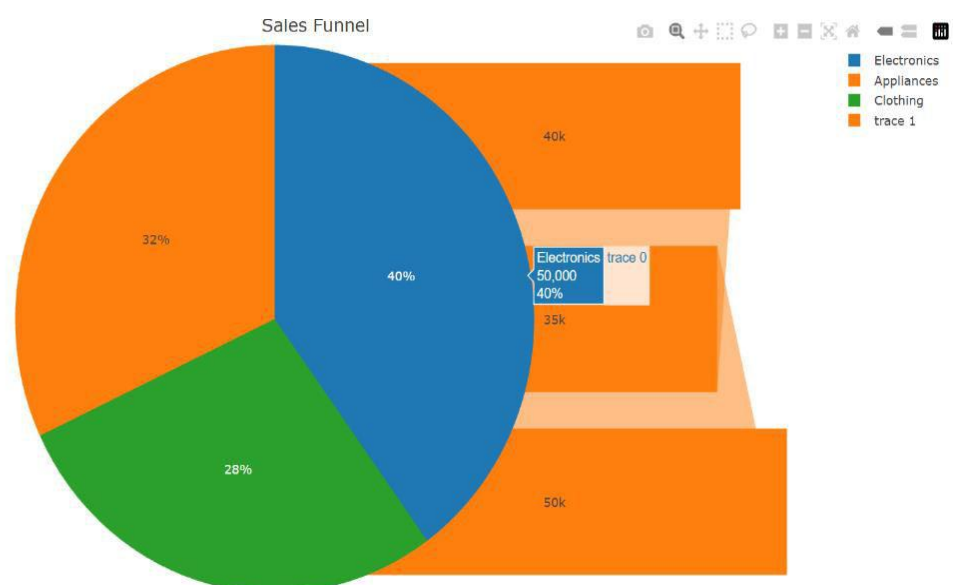
> |

11 C. Interactive Dashboard (R)

Program:

```
pie_plot <- plot_ly(  
  product_data,  
  labels=~Category,  
  values=~Sales,  
  type="pie") %>%  
  layout(title="Sales Distribution")  
  
funnel_plot <- plot_ly(  
  product_data,  
  type="funnel",  
  y=~Category,  
  x=~Sales) %>%  
  layout(title="Sales Funnel")  
  
dashboard <- subplot(  
  pie_plot,  
  funnel_plot,  
  nrows=1)  
dashboard
```

Output:



11 D. Pie Chart

Program:

```
ggplot(product_data,  
aes(x="",  
y=Sales,  
fill=Category))+  
geom_bar(stat="identity",width=1)+  
coord_polar("y")+  
labs(title="Sales Distribution Across Product Categories")+  
theme_void()
```

Output:

Sales Distribution Across Product Categories



SET 12- Website Traffic

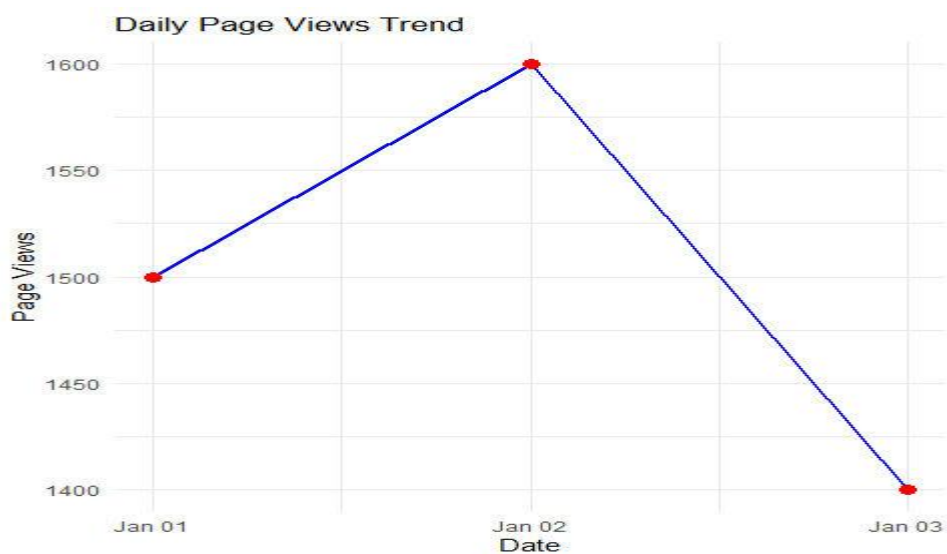
12 A. Line Chart

Program:

```
website_data <- data.frame(  
Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),  
PageViews=c(1500,1600,1400),  
CTR=c(2.3,2.7,2.0))  
ggplot(website_data,  
aes(x=Date,y=PageViews,group=1))+
```

```
geom_line(color="blue",size=1)+
geom_point(color="red",size=3)+
labs(title="Daily Page Views Trend",
x="Date",
y="Page Views")+
theme_minimal()
```

Output:

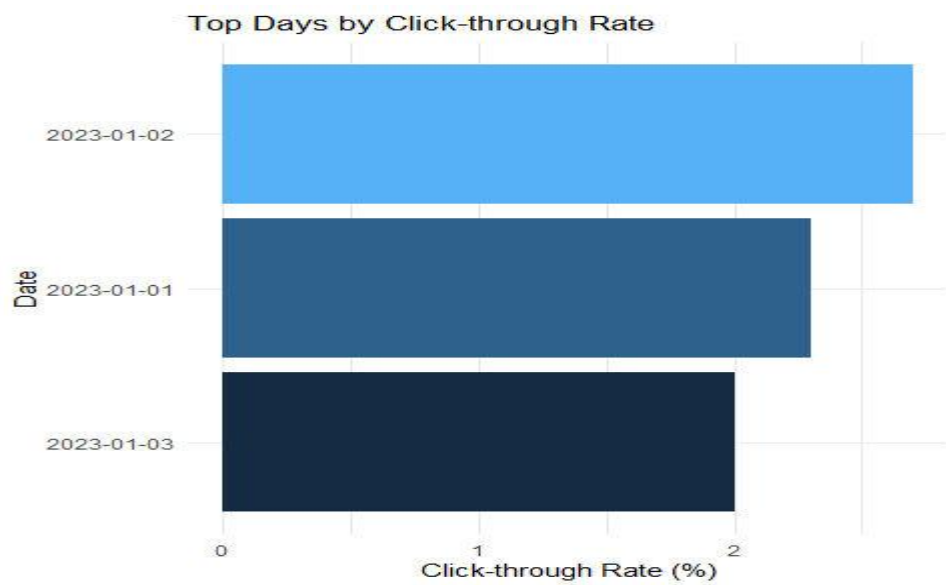


12 B. Bar Chart (Top Days by Click-through Rate)

Program:

```
top_ctr <- website_data[order(-website_data$CTR),]
ggplot(top_ctr,
aes(x=reorder(as.character(Date),CTR),
y=CTR,
fill=CTR))+
geom_bar(stat="identity")+
coord_flip()+
labs(title="Top Days by Click-through Rate",
x="Date",
y="Click-through Rate (%)")+
theme_minimal()+
theme(legend.position="none")
```

Output:



12 C. Table

Program:

```
kable(  
  website_data,  
  caption="Website Traffic Data")
```

Output:

Table: Website Traffic Data

Date	PageViews	CTR
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0

12 D. Interactive Dashboard

Program:

```
line_plot <- ggplotly(  
  ggplot(website_data,  
    aes(x=Date,y=PageViews,group=1))+  
  geom_line(color="blue")+  
  geom_point(color="red",size=3)+  
  labs(title="Daily Page Views"))
```

```

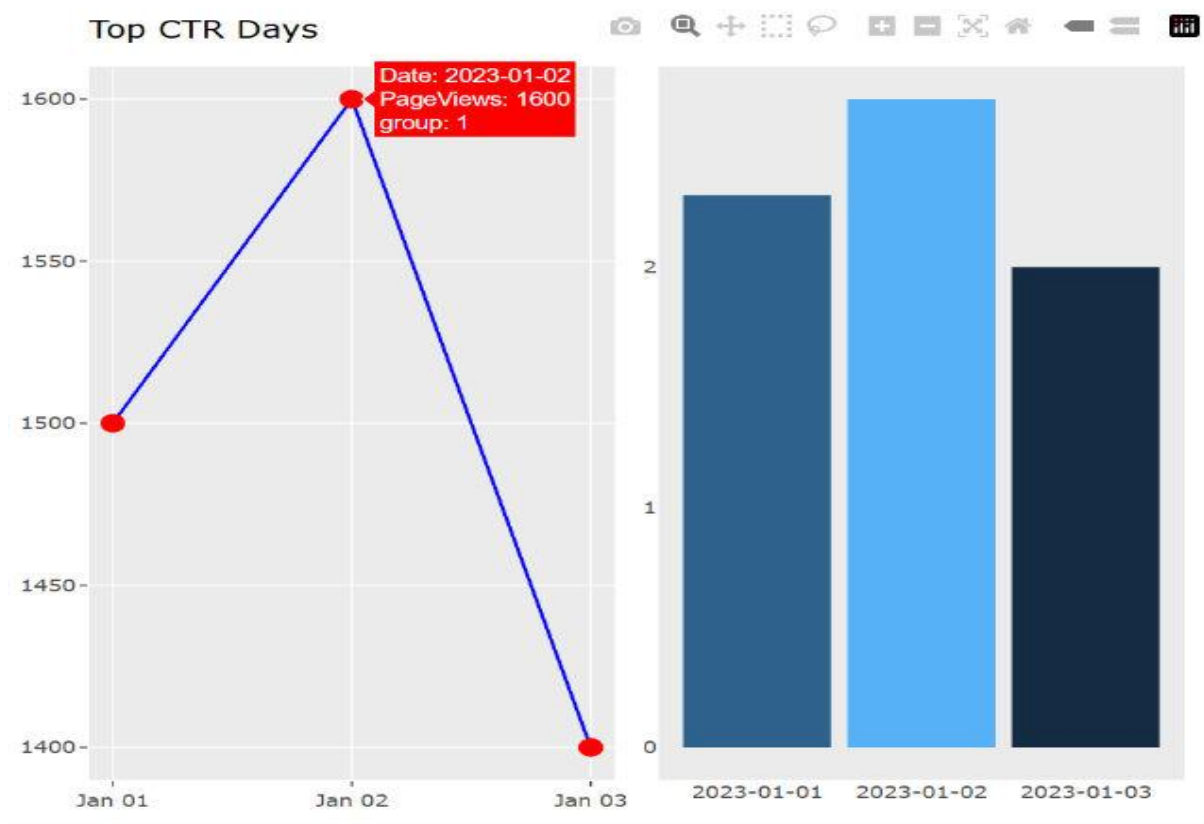
bar_plot <- ggplotly(
  ggplot(top_ctr,
    aes(x=as.character(Date),
      y=CTR, fill=CTR))+
    geom_bar(stat="identity")+
    labs(title="Top CTR Days")+
    theme_minimal()+
    theme(legend.position="none"))

dashboard <- subplot(
  line_plot,
  bar_plot, nrow=1,
  shareX=FALSE,
  shareY=FALSE)

dashboard

```

Output:



SET 13-Geographic Data

13 A. Map Chart

Program:

```
import pandas as pd
import plotly.express as px

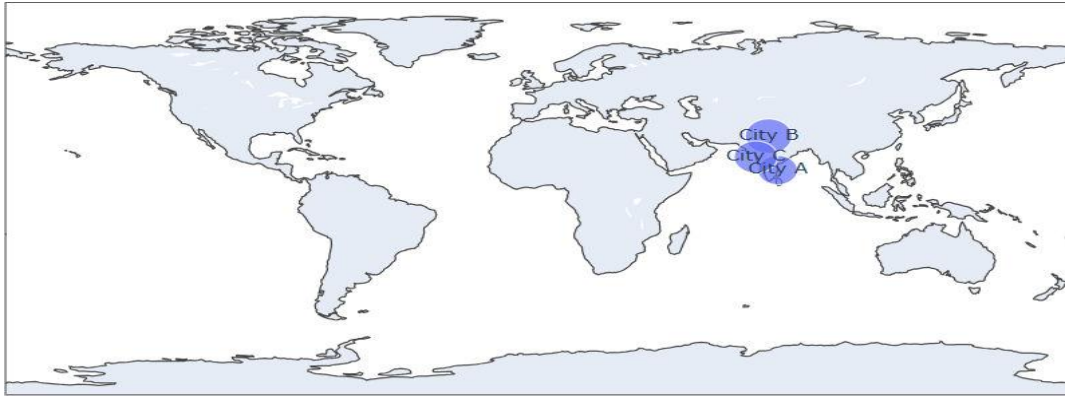
data = pd.DataFrame({
    "City":["City A","City B","City C"],
    "Population":[500000,700000,600000],
    "Temperature":[75,68,80],
    "Elevation":[1000,800,1200],
    "Latitude":[13.08,28.61,19.07],
    "Longitude":[80.27,77.21,72.87]
})

fig = px.scatter_geo(
    data,
    lat="Latitude",
    lon="Longitude",
    text="City",
    size="Population",
    title="Distribution of Cities"
)

fig.update_layout(
    geo=dict(showland=True),
    title_x=0.5
)

fig.show()
```

Output:



13 B. Scatter plot

Program:

```
import plotly.express as px

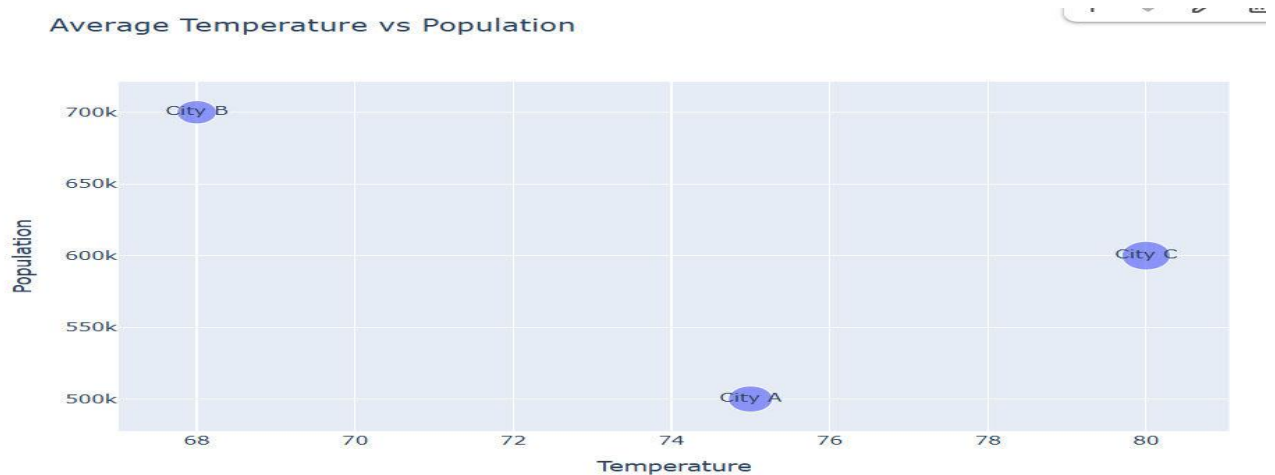
fig = px.scatter(
    data,
    x="Temperature",
    y="Population",
    text="City",
    size="Elevation",
    title="Average Temperature vs Population"
)

fig.show()

print("Insight:")

print("City B has the highest population with the lowest temperature, while City C has the highest temperature.")
```

Output:



13 C. Table

Program:

```
from IPython.display import display
display(data)
```

Output:

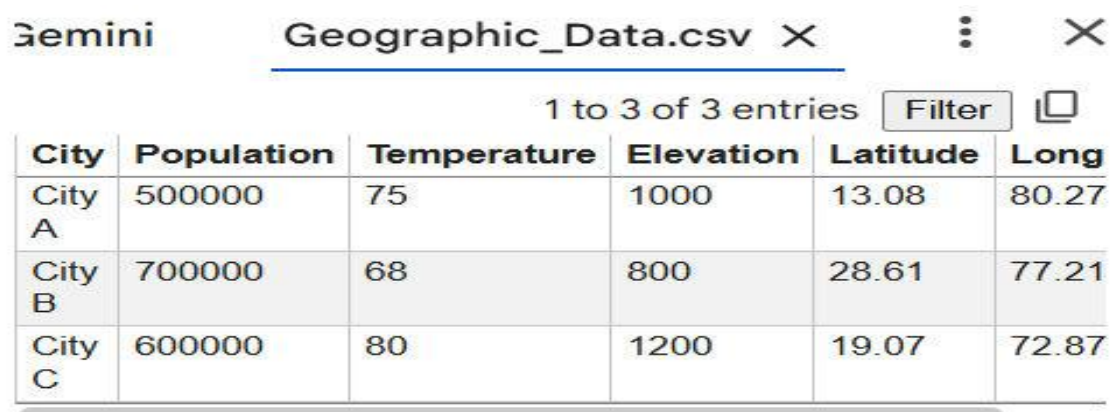
	City	Population	Temperature	Elevation	Latitude	Longitude
0	City A	500000	75	1000	13.08	80.27
1	City B	700000	68	800	28.61	77.21
2	City C	600000	80	1200	19.07	72.87

13 D. Tableau Dashboard

Program:

```
data.to_csv("Geographic_Data.csv", index=False)
```

Output:



The screenshot shows a Tableau interface with a window titled "Geographic_Data.csv". The table displays 3 entries. The columns are City, Population, Temperature, Elevation, Latitude, and Long. The data rows are: City A (Population: 500000, Temperature: 75, Elevation: 1000, Latitude: 13.08, Long: 80.27), City B (Population: 700000, Temperature: 68, Elevation: 800, Latitude: 28.61, Long: 77.21), and City C (Population: 600000, Temperature: 80, Elevation: 1200, Latitude: 19.07, Long: 72.87). The interface includes a "Filter" button and a "1 to 3 of 3 entries" indicator.

City	Population	Temperature	Elevation	Latitude	Long
City A	500000	75	1000	13.08	80.27
City B	700000	68	800	28.61	77.21
City C	600000	80	1200	19.07	72.87

SET 14-Survey Results

14 A. Stacked Bar Chart

Program:

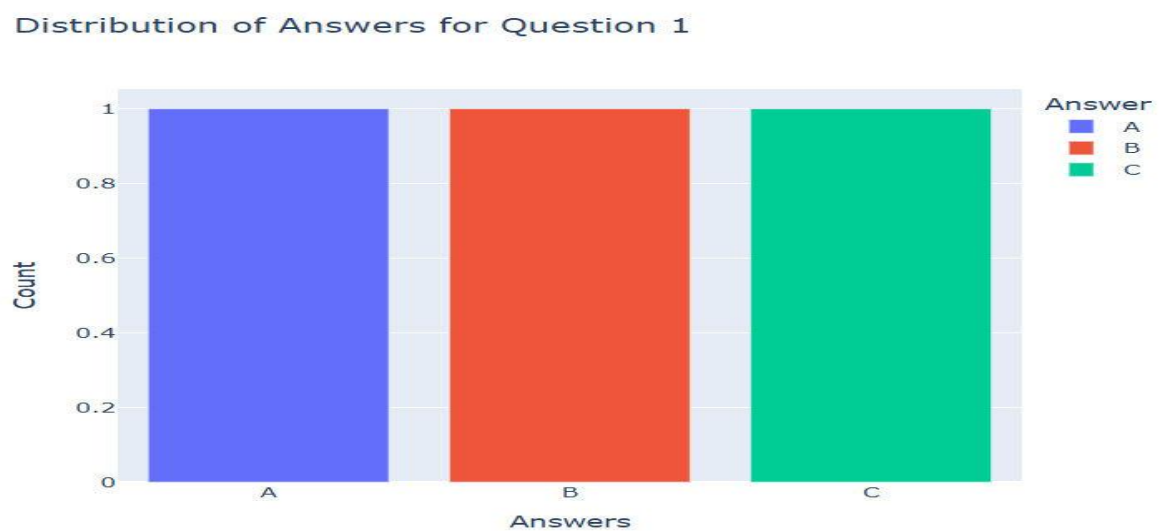
```
import pandas as pd
import plotly.express as px
survey = pd.DataFrame({
    "Respondent": [1, 2, 3],
    "Question 1": ["A", "B", "C"],
    "Question 2": ["B", "A", "A"],
    "Question 3": ["C", "D", "B"]
})
```

```

})
count = survey["Question 1"].value_counts().reset_index()
count.columns = ["Answer", "Count"]
fig = px.bar(
    count,
    x="Answer",
    y="Count",
    color="Answer",
    title="Distribution of Answers for Question 1"
)
fig.update_layout(
    xaxis_title="Answers",
    yaxis_title="Count"
)
fig.show()

```

Output:



14 B. Radar chart

Program:

```

import plotly.graph_objects as go
responses = [
    survey["Question 1"].nunique(),

```

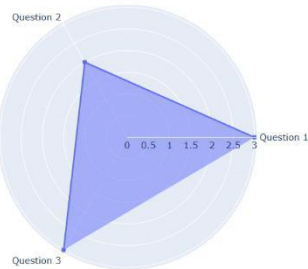
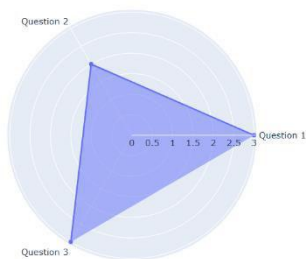
```

survey["Question 2"].nunique(),
survey["Question 3"].nunique()
]
fig = go.Figure()
fig.add_trace(go.Scatterpolar(
    r=responses,
    theta=["Question 1","Question 2","Question 3"],
    fill='toself',
    name='Responses'
))
fig.update_layout(
    polar=dict(radialaxis=dict(visible=True)),
    title="Survey Response Pattern"
)
fig.show()

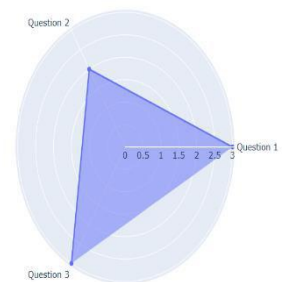
```

Output:

Survey Response Pattern



Survey Response Pattern



14 C. Table

Program:

```

from IPython.display import display
display(survey)

```

Output:

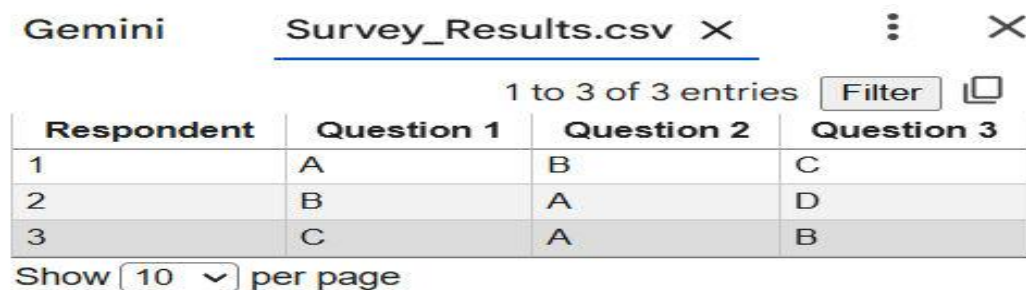
	Respondent	Question 1	Question 2	Question 3
0	1	A	B	C
1	2	B	A	D
2	3	C	A	B

14 D. Tableau Dashboard

Program:

```
survey.to_csv("Survey_Results.csv", index=False)
```

Output:



The screenshot shows a Tableau interface with a dashboard titled "Survey_Results.csv". The dashboard displays a table with 3 entries. The table has four columns: "Respondent", "Question 1", "Question 2", and "Question 3". The data is as follows:

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Below the table, there is a "Show 10 per page" dropdown menu.

SET 15-Product Sales Analysis

15 A. Grouped Bar Chart

Program:

```
import pandas as pd
import plotly.express as px

sales = pd.DataFrame({
    "Product":["Product A","Product B","Product C"],
    "January":[2000,1500,1200],
    "February":[2200,1800,1400],
    "March":[2400,1600,1100]
})

sales_long = sales.melt(id_vars="Product",
    var_name="Month",
    value_name="Sales")

fig = px.bar(
    sales_long,
    x="Product",
    y="Sales",
    color="Month",
    barmode="group",
    title="Monthly Product Sales - First Quarter"
```

```
)
fig.update_layout(
    xaxis_title="Product",
    yaxis_title="Sales"
)
fig.show()
```

Output:

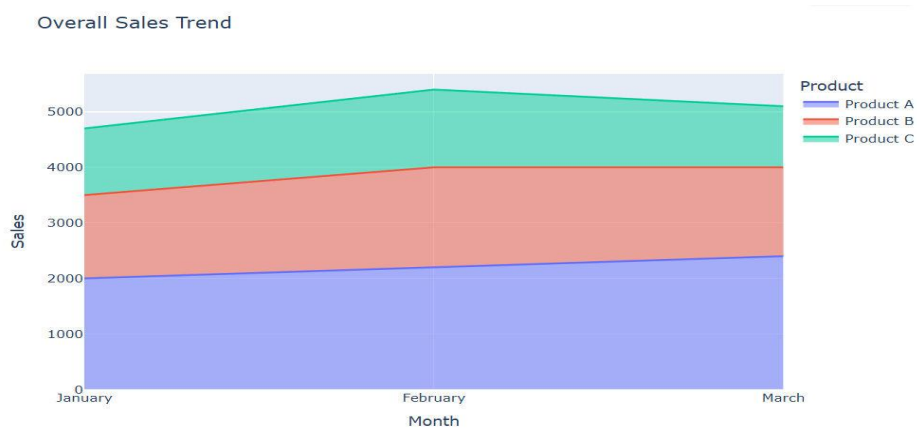


15 B. Stacked Area Chart

Program:

```
fig = px.area(
    sales_long,
    x="Month",
    y="Sales",
    color="Product",
    title="Overall Sales Trend"
)
fig.update_layout(
    xaxis_title="Month",
    yaxis_title="Sales"
)
fig.show()
```

Output:



15 C. Table

Program:

```
from IPython.display import display
display(sales)
```

Output:

	Product	January	February	March
0	Product A	2000	2200	2400
1	Product B	1500	1800	1600
2	Product C	1200	1400	1100

15 D. Tableau Dashboard

Program:

```
sales.to_csv("Monthly_Product_Sales.csv", index=False)
```

Output:

Monthly_Product_Sales.csv X Gemi ⋮ X

1 to 3 of 3 entries Filter

Product	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

Show 10 per page